MIL-P-60258 (MU)
30 November 1965
SUPERSEDING
LCA-PD-50, Rev. B
5 November 1965

MILITARY SPECIFICATION

PROJECTILE, 20MM, TARGET PRACTICE, M55A2B1

1. SCOPE

1.1 This specification covers one model projectile for Cartridges, 20MM, TP, M55A2 for use in Guns, Automatic, 20MM, M39 and M61.

2. APPLICABLE DOCUMENTS

The following documents of the issue in effect on date of invitation for bids, or a request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning Methods, and Pretreatment of Ferrous Surface for Organic Coatings

MILITARY

MIL-P-10025 - Packing and Marking for Domestic Shipment of Inert Ammunition Components, General Specification For

STANDARDS

FEDERAL

Federal Test Methods Standard No. 151 - Metals; Test Methods.

FSC 1305

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109 - Quality Assurance Terms and Definitions

MIL-STD-651 - Visual Inspection Standards for 20MM Ammunition and Components

MIL-STD-1167 - Military Standard Ammunition Data Card

MIL-STD-1235(ORD) - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

MUNITIONS COMMAND

D7259126 - Projectile, 20MM Target Practice M55A2B1
D7259127 - Projectile, Unpainted
D7259128 - Projectile, Unbanded
D7259129 - Body
B7258816 - Blank, Rotating Band, for Projectiles, 20MM
F7258861 - Carton for Interplant Shipment of Projectiles, 20MM, Assemblies and Metal Parts Assemblies (200)
D7259191 - Cartridge, 20MM for M55A2, TP & M55A2B1, TP, Projectile Metal Parts Test
D7259136 - Shipping Container for Projectile, 20MM, M55A2 Assemblies (10,000)

PUBLICATIONS

MUNITIONS COMMAND

AMCR 715-505 - Ammunition Ballistic Acceptance Test Methods, Test Procedures for 20MM Cartridges

IEL 7259126 - Index of Inspection Equipment Lists for: Projectile, 20MM, Target Practice, M55A2B1

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 General.-Materials and the projectile shall comply with the requirements of applicable drawings and specifications, and with all other requirements specified herein.

3.2 First article testing.-Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.2.1 Manufacturing process.-All parts and assemblies shall be manufactured by a process approved by the contracting officer, and no deviation from that process shall be made without his prior approval.

3.3 Banding.-Prior to application of the rotating band, the band seat shall be cleaned to remove all foreign matter. Shot or sand blasting is not permitted for cleaning the band seat after knurling. The rotating band blank shall be pressed to obtain band seat fill without radial void in excess of 0.004 inch, average.

3.4 Soundness.-The projectile shall be free of ferrous metal discontinuity to the extent detectable by ultrasonic method of inspection, or other method approved by the contracting officer.

3.5 Protective coating.

3.5.1 Phosphate coating.-The pretreatment process used shall conform to the requirements of TT-C-490 for Type I, zinc phosphate. The phosphate coating deposited on the projectile body by the pretreatment process, shall comply with the requirements of that specification for texture, continuity, appearance, cleanliness and minimum weight.

3.5.2 Paint coating.-The prescribed enamel or lacquer shall be applied to the pretreated projectile in a manner to produce a coating of uniform thickness over the entire specified areas, with adhesion as required by TT-C-490.

3.5.3 Corrosion resistance.-The projectile shall show no visible evidence of ferrous metal corrosion after exposure to a 5-percent salt solution applied for 24 hours, as prescribed by Method 811.1 of Federal Test Methods Standard No. 151.

3.6 Metal parts security.-The projectile shall show no evidence of separation or breakup of metal part(s) when assembled in accordance with Dwg. D7259191, and tested for metal parts security.

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3.7 Workmanship.-The projectile and its parts shall be free of defects such as metal flaws, nicks, burrs, cuts, foreign matter, and marking and packing deficiencies.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms.

4.2 First article sample.-At the beginning of regular production a sample shall be submitted in accordance with contract requirements and shall consist of:

a. For physical and dimensional inspection and tests.

(1) 100 projectiles (painted).

(2) 100 projectiles (prior to pretreatment for painting).

(3) 100 bodies.

(4) 100 projectiles (unpainted, unbanded).

(5) 25 rotating band blanks.

b. For ballistic security test.

4,000 projectiles (painted).

4.2.1 Examination and test.-After inspection and provisional acceptance by the Government, the sample shall be inspected for all the requirements of the drawings and specifications. The ballistic security portion of the sample will not be assembled and tested for metal parts security until all other inspection and test requirements have been met.

4.2.2 First article sample failure.-Failure of the sample to meet the requirements of the drawings and specifications shall result in sample rejection.

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4.3 Examination.

4.3.1 Lot formation.-A production lot shall consist of the projectiles produced by one manufacturer in one unchanged process, in accordance with the same drawings and drawing revisions, and the same specification and specification revision.

4.3.2 Critical defects.-Each projectile, prior to preparation for painting, shall be examined by the contractor for ferrous metal discontinuity, using the method specified in 4.5.2, or other method approved by the contracting officer, and each projectile shall be examined for the critical defect of 4.3.4.5. All defective projectiles shall be rejected.

4.3.3 Sampling and acceptance criteria.-Sampling and acceptance criteria shall be in accordance with MIL-STD-105 or MIL-STD-1235, as appropriate. Acceptance shall be based upon the individual defect characteristic or upon classes of defect characteristics with individual defect limitation on major defects. When inspection is performed on a major class of defects, no individual defect of the class shall exceed the acceptance number for the AQL assigned to individual defects associated with that class, i.e., the number of individual major defects shall be limited to less than that permitted for the class (see 6.2).

4.3.3.1 AQL's.-The following AQL's are assigned:

a. Individual defect characteristic basis.

Major defects	0.25%
Minor defects	0.40%

b. Defect characteristic, class basis.

Major defects	1.50%
Minor defects	2.50%

c. Individual defect characteristic associated with the class basis.

Major defects	0.40%

4.3.4 Classification of defects.

4.3.4.1 Rotating band blank (see Dwg. B7258816).

Categories and defects	Method of inspection
Critical	
None defined.	

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Major

101.	Foreign matter	Visual
102.	Defective metal (porosity, lamination, or inclusion)	Visual
103.	Width not full for 360°	Visual (see 4.3.5.2)

Minor

201.	Width, max.	Gage

4.3.4.2 Body (see Dwg. D7259129).

Categories and defects	Method of inspection

Critical

None defined.

Major

101.	Thickness of base	Gage
102.	Inside diameter, max	Gage
103.	Wall thickness variation	Gage

Minor

None defined.

4.3.4.3 Projectile unbanded (see Dwg. D7259128).

Categories and defects	Method of inspection

Critical

None defined.

Major

101.	Knurling missing or incomplete	Visual
102.	Foreign matter in rotating band seat	Visual
103.	Length, nose to rear rotating band seat	Gage
104.	Depth, knurling, min.	Visual-Gage (see 4.3.5.1)
105.	Width, knurling, min.	Visual

Minor

201.	Diameter, rotating band seat	Gage
202.	Width, rotating band seat	Gage

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4.3.4.4 Projectile, unpainted (see Dwg. D7259127).

Categories and defects		Method of inspectio
Critical		
	None defined.	
Major		
101.	Length, nose to rear edge of rotating band	Gage
102.	Over-all length	Gage
103.	Concentricity, rotating band with bourrelet	Gage
104.	Diameter, bourrelet, min.	Gage
105.	Diameter, body, rear of rotating band	Gage
106.	Diameter, rotating band	Gage
107.	Concentricity, body rear of rotating band, with bourrelet	Gage
108.	Width, rotating band	Gage
109.	Location, crimping groove	Gage
110.	Depth, crimping groove	Gage
111.	Radius, crimping groove	Gage
112.	Radius, rear edge of crimping groove	Gage
113.	Concentricity of nose with bourrelet	Gage
114.	Profile of ogive and nose (see Dwg. D7259128)	Gage
115.	Unfilled band seat (gap, except radial gap)	Visual
116.	Damage, except rotating band	Visual-Feel
117.	Unfilled rotating band seat (gap)	Visual
Minor		
201.	Finish improper	Visual
202.	Depth and width of undercut, front and rear of rotating band	Gage
203.	Base radius missing	Visual

4.3.4.5 Projectile, 20mm, target practice, M55A2B1 (see Dwg. D7259126).

Categories and defects		Method of inspection
Critical		
1.	Hole, void or crack in nose apex	Visual
Major		
101.	Diameter, bourrelet, max.	Gage
102.	Diameter, body rear of rotating band, max.	Gage
103.	Weight	Balance
104.	Foreign matter	Visual

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Minor

201.	Exterior paint not dry (when packed)	Manual
202.	Marking incorrect, incomplete, illegible, or missing	Visual
203.	Paint on rotating band	Visual
204.	Rotating band damaged	Visual
205.	Paint inadequate	Visual

4.3.5 Definition of defects.-MIL-STD-651 will be used as a guide for grading visual defects.

4.3.5.1 A rotating band seat which has less than minimum depth knurl shall be classed defective. Visual examination may be used to determine depth of knurl, with gaging to apply when the depth is visually suspect or borderline.

4.3.5.2 A rotating band blank judged to lack sufficient metal in any part of its width or thickness to provide for acceptable band seating and finished profile dimensions, shall be classed defective.

4.4 Tests.-Each lot of projectiles shall be tested for the quality characteristics listed in Table I. The metal parts security test will be performed by Government personnel at a facility designated by the procurement activity. Test quantities and acceptance criteria shall be as indicated. The lot of projectiles shall be rejected if the sample of any test fails to meet the specified acceptance criteria. Test methods and procedures shall be as specified in 4.5.

TABLE I

Test	Item and sample size
Band tightness	Projectiles, unpainted. MIL-STD-105, Code N, normal.
Soundness verification	300 projectiles prior to rotating band seat machining.
Phosphate coating weight	One test panel at the beginning of daily production and at four hour intervals thereafter.
Paint adhesion	Projectiles, painted. MIL-STD-105, Code F, normal.

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TABLE I (Cont)

Test	Item and sample size
Corrosion resistance	Projectiles, painted. MIL-STD-105, Code H, normal.
Metal parts security	250 projectiles, painted.

4.4.1 Band tightness.-A double sampling plan using the method of MIL-STD-105 shall be used in this test, applying an AQL of 0.10%. A radial gap indication exceeding 0.004 inch shall be classed as a defect.

4.4.2 Soundness verification.-The acceptance number associated with this sample is zero (0). Unsound ferrous metal of the degree defined in 4.5.2.1 shall be classed as a defect.

4.4.3 Phosphate coating weight.-Failure of the standard panel to meet the acceptance criteria of TT-C-490 for phosphate coating weight, texture, continuity, appearance and cleanliness, shall cause rejection of all projectiles phosphated since the preceding test.

4.4.4 Paint adhesion.-A double sampling plan using the method of MIL-STD-105 shall be used in this test applying an AQL of 2.50%. Failure of any sample projectile to meet the requirements for paint adhesion shall be classed as a defect.

4.4.5 Corrosion resistance.-A double sampling plan using the method of MIL-STD-105 shall be used in this test, applying an AQL of 1.0%. Evidence of red rust on the exterior surface of any sample projectile forward of the rotating band shall be classed as a defect.

4.4.6 Metal parts security.-The lot will be rejected, not subject to retest, if any one of the following types of metal parts separations occurs (as evidenced by screen indication or recovered fragment(s):

a. Projectile breakup.

b. Complete band separation.

Complete band separation is defined as the entire band or part of the band having band seat serration imprint becoming detached, as evidenced by the recovered piece(s) or hole(s) in the screen. The lot will not be penalized for normal band fringing. Band fragment(s) showing no evidence of band seat serration imprint and having length not greater than 0.500 inch will be treated as follows:

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(1) The lot will be rejected, not subject to retest, if three or more assemblies throw fragments.

(2) The lot will be accepted if two or less projectiles throw fragments. In the case of fragments greater than 0.500 inch, the disposition of the lot will be determined by the procuring activity.

4.4.7 Packing and marking inspection.-Inspection for packing and marking inspection shall be as prescribed by MIL-P-10025, as applicable to the packing drawing.

4.4.8 Inspection equipment.-The examinations and tests shall be made using the equipment listed on IEL 7259126.

4.5 Test methods.

4.5.1 Band tightness.-The test for band tightness shall be conducted using the band tightness tester as shown on IEL 7259126. The tester shall have an initial ram pressure of 200 p.s.i., and a final ram pressure of 3,000 p.s.i. Inasmuch as four indentors are used, the average radial gap measurement applicable to band tightness shall be one-fourth of that shown on the tester indicator.

4.5.2 Soundness.-The test of body soundness shall be performed by an ultrasonic method approved by the procuring activity, that introduces a shear angle ultrasonic beam into the projectile submerged in water. The test apparatus shall be adjusted to inspect projectiles for defects in the zones specified on Dwg. D7259128. The inspection equipment and procedure shall be developed by the contractor, and shall have technical approval prior to application.

4.5.2.1 Soundness verification.-A projectile having a flaw indication when tested using the method of 4.5.2, shall be sectioned normal to and at the approximate midpoint of the indication. The sectioned specimen will be etched, polished and examined at 100 diameters. An indication which is shown to be ferrous metal discontinuity in magnitude greater than .005 inch in radial from the exterior surface depth, will cause the projectile to be classed defective. This test will be conducted by Government personnel.

4.5.3 Phosphate coating.-The method of test shall be as prescribed by TT-C-490, using standard panel(s) of specified material and dimensions. The texture, continuity, appearance and cleanliness of the coating shall be recorded along with the coating weight.

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4.5.4 Paint adhesion.-The method of test shall be as prescribed by TT-C-490.

4.5.5 Corrosion resistance.-The test of corrosion resistance by exposure to salt spray shall be conducted in accordance with Method 811.1 of Federal Test Methods Standard No. 151 - Metals, Test Methods. The base end of the sample projectiles, including the rotating band, may be masked out by a wax coating or other suitable material.

4.5.6 Metal parts security.-The sample projectiles (painted) will be assembled in accordance with Dwg. D7259191, Cartridge, 20mm, for projectile metal parts test. The method of test will be as prescribed in AMCR 715-505, Volume 8. The test will be fired in bursts of fifty rounds in an M39 gun. The gun barrel will be at ambient temperature for the firing of each burst (see 6.3).

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level C.-The projectiles shall be packed in accordance with Dwg. F7258861 or D7259136, as specified by the procuring agency.

5.2 Marking.

5.2.1 Level C.-Marking on packs shall be in accordance with specification MIL-P-10025.

5.3 Data cards.-The material size and format of data cards shall be in accordance with MIL-STD-1167. When data cards are specified in the contract, they shall be prepared and distributed by the contractor.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:

a. Title, number, and date of this purchase description.

b. Place of inspection, if not place of manufacture.

c. Instructions for shipment of production lot samples for metal parts security test.

d. Detailed packing instructions (see 5.1).

e. Data card requirements (see 5.3).

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f. Provision for the supply, maintenance, and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.2 AQL's.-The optional use of AQL values for either individual defects or classes of defects, with individual major defect limitations, is intended to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also provides flexibility where sampling inspection for acceptance is integrated into the manufacturing process. When three or less major defects are listed in a classification of defects, inspection by class of defects shall not be used.

6.3 Proving ground notification.-The contracting officer will notify the proving ground when to proceed with the first article ballistic security test (see 4.2.1).

6.4 Inspection equipment.-Design status of all inspection equipment necessary to perform the examination and tests prescribed in this purchase description is indicated in IEL 7259126. Changes to the design of equipment contained in the IEL, when the inspection equipment is used for acceptance for, or by, the Government, shall require approval by Lake City Army Ammunition Plant, Industrial Engineering Division, Independence, Missouri.

Custodian:

Army - MU

Preparing Activity:

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Project No. 1305-A374

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